

Curriculum vitae

Feiyu Quan (全飞宇)

feiyu.quan@link.cuhk.edu.hk | [Website](#) | [Google Scholar](#) | [ORCID](#) Last updated: July 19, 2026

Education

The Chinese University of Hong Kong

Doctor of Philosophy, Astrophysics

2024 – Present

Supervisor: [Hua-bai Li](#) (李華白)

University of Cambridge

Master of Advanced Study, Part III of the Mathematical Tripos

2023 – 2024

Essay: Structure Formation in Saturn's Rings, with [Henrik Latter](#)

University of Toronto

Honours Bachelor of Science (with High Distinction), Physics and Astrophysics

2019 – 2023

Publications

- **F. Quan** et al., Projection Is All You Need: Interpreting Polarization Measurements in the Orion Clouds with Sub-Alfvénic MHD Simulations, accepted to ApJ.
- J. E. Pineda et al. (including **F. Quan**), The Green Bank Ammonia Survey: Data Release 2, *The Astrophysical Journal Supplement*, 2026, <https://doi.org/10.3847/1538-4365/ae11b1>
- A. Pandhi et al. (including **F. Quan**), Alignment of dense molecular core morphology and velocity gradients with ambient magnetic fields, *Monthly Notices of the Royal Astronomical Society*, 2023, <https://doi.org/10.1093/mnras/stad2283>

Selected Research Experience

Research Project: New algorithm to detect resonances in planetary systems

Fall 2022

- We aim to improve an existing method for detecting mean motion resonances in planetary system. The method may be applied to secular resonances as well.

Supervisor: [Hanno Rein](#)

CITA Research Fellowship: How do disk galaxies warp?

Summer 2022

- This project at Canadian Institute for Theoretical Astrophysics explores the processes that drive disk deformations, by analyzing outputs from Illustris TNG50, a state-of-the-art magneto-hydrodynamical cosmological simulation that computes the formation and evolution of hundreds of thousands of galaxies.

Supervisor: [Neige Frankel](#) and [Joshua Speagle](#)

Research Assistant: Investigate the statistical properties of star-forming clouds

Fall 2021, Winter 2022

- The project investigates the statistical properties of star-forming clouds in our Galaxy, including gas densities, temperatures, and velocity structures, to compare these observed values with predictions from numerical simulations of how stars form. Financially supported by the University of Toronto Work Study program.

Supervisor: [Rachel Friesen](#)

Research Project: Dynamical analysis of a recently discovered extra-solar planetary system using numerical tools Summer 2021

- Analysis of planetary system TOI-1260. We applied numerical simulation to the system to study its dynamical stability, and used MCMC to constrain the rate of precession of planet orbit to predict their transits.

Supervisor: [Hanno Rein](#)

Teaching Experience

Teaching Assistant: UGEB2401, Astronomy 2024, 2025, 2026

- Conducted a 2 hour consultation session each week for students including working through problems and demonstrating key problem solving methods. Managing detailed record of students performance.

Teaching Assistant: PHYA10, Physics I for the Physical Sciences 2021

- Facilitating a 2 hour practical session each week for approximately 20 students including working through problems in small groups and demonstrating key problem solving methods. Managing detailed record of students performance.

Physics Tutor at Environmental and Physical Sciences Students Association 2020 – 2021

- Physics tutor at Environmental and Physical Sciences Students Association (EPSA). Helping students understand concepts and solve problems in first-year physics (PHYA10/A11, PHYA21/A22).

Skills and Interests

Familiar with Python, Jupyter, $\text{\LaTeX}$ and C. Basic knowledge of Linux, Git, and Fortran. Interested in applying numerical simulations and techniques to solve various problems in astrophysics.

Awards

Hong Kong PhD Fellowship Scheme	2024 – 2028
CUHK Vice-Chancellor's Scholarship	2024
University of Toronto Dean's List	2020 – 2023
Canadian Institute for Theoretical Astrophysics Research Fellowship	2022
Department Excellence and Leadership Award	2022